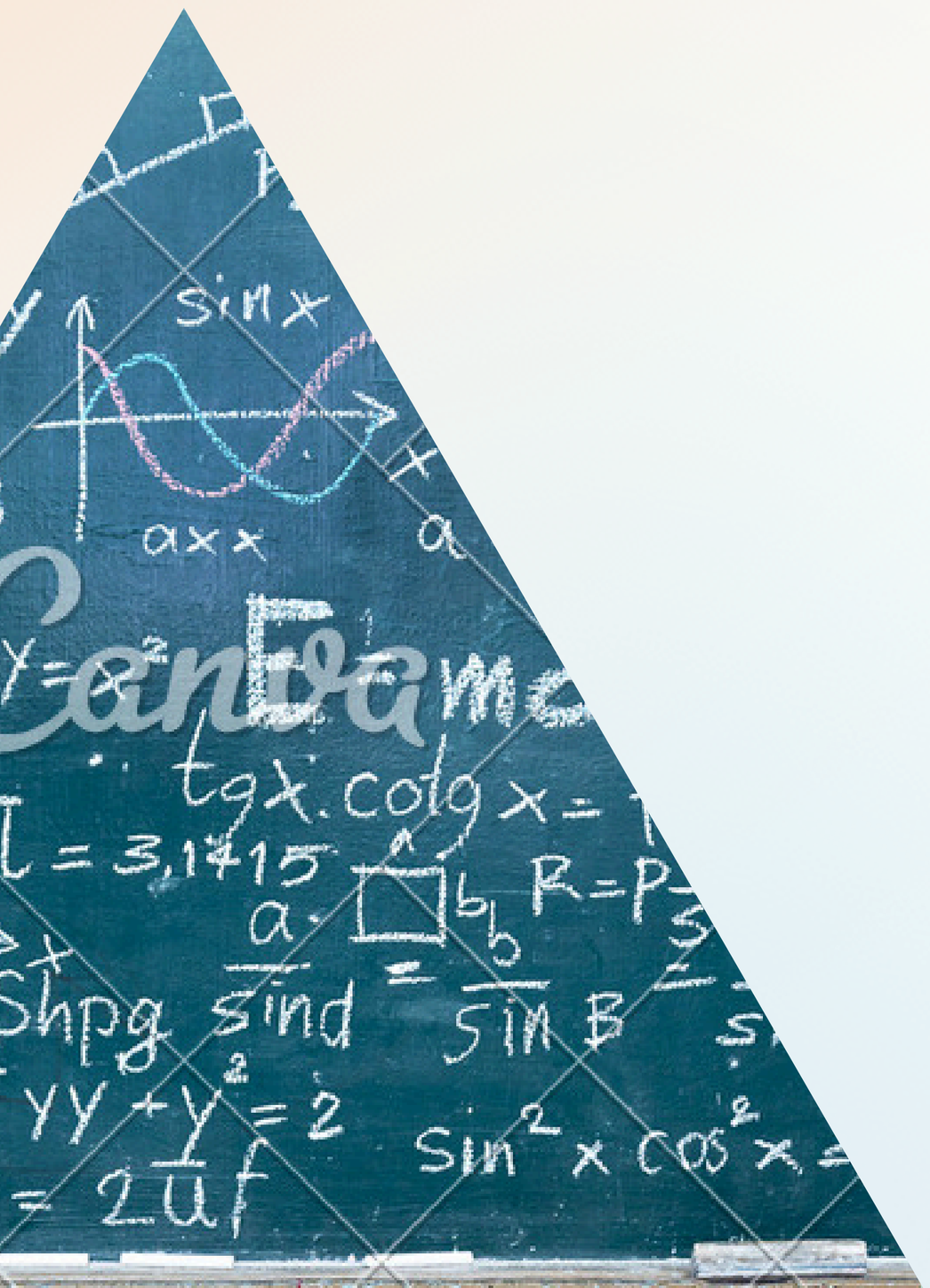


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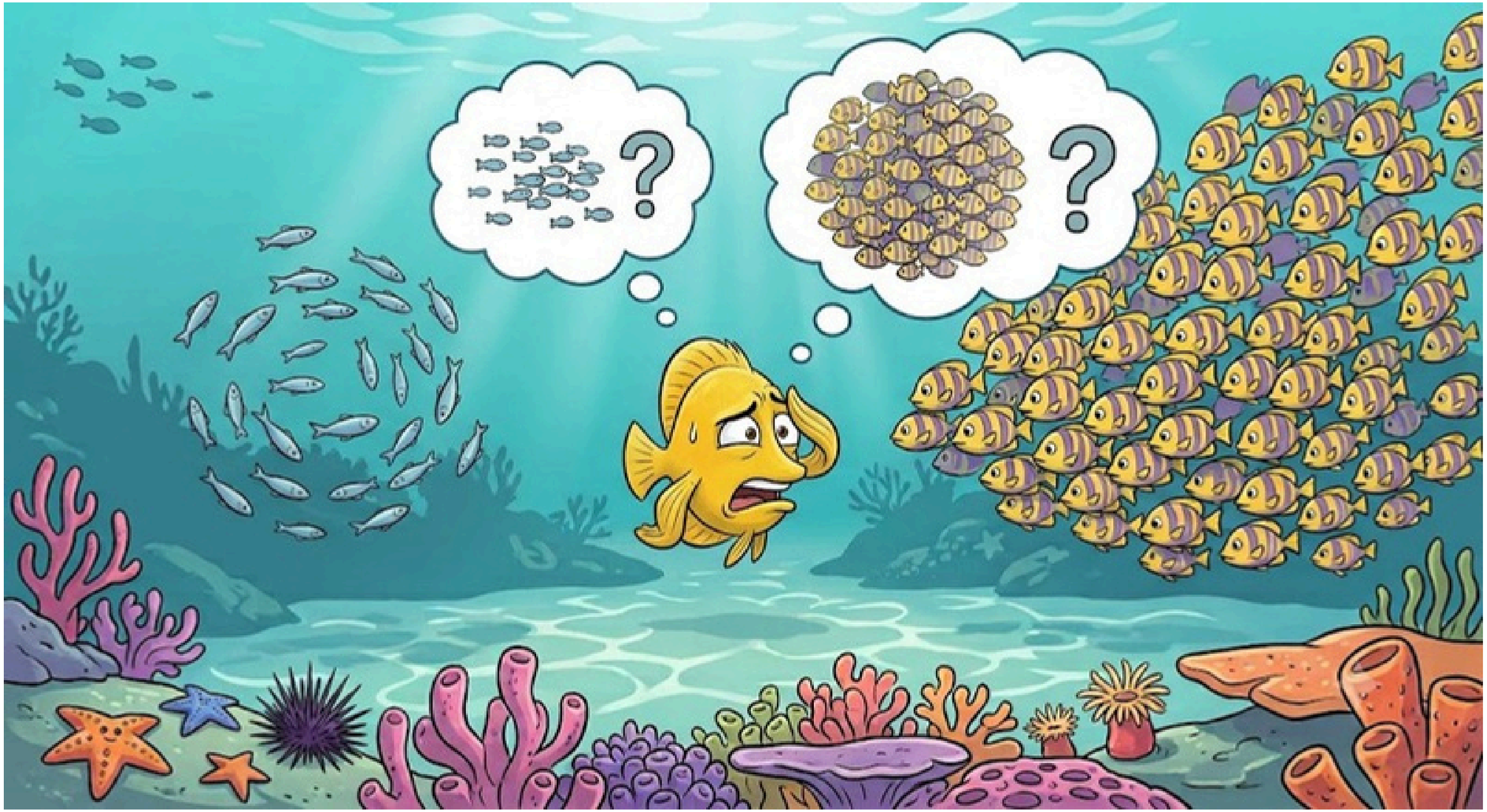
8

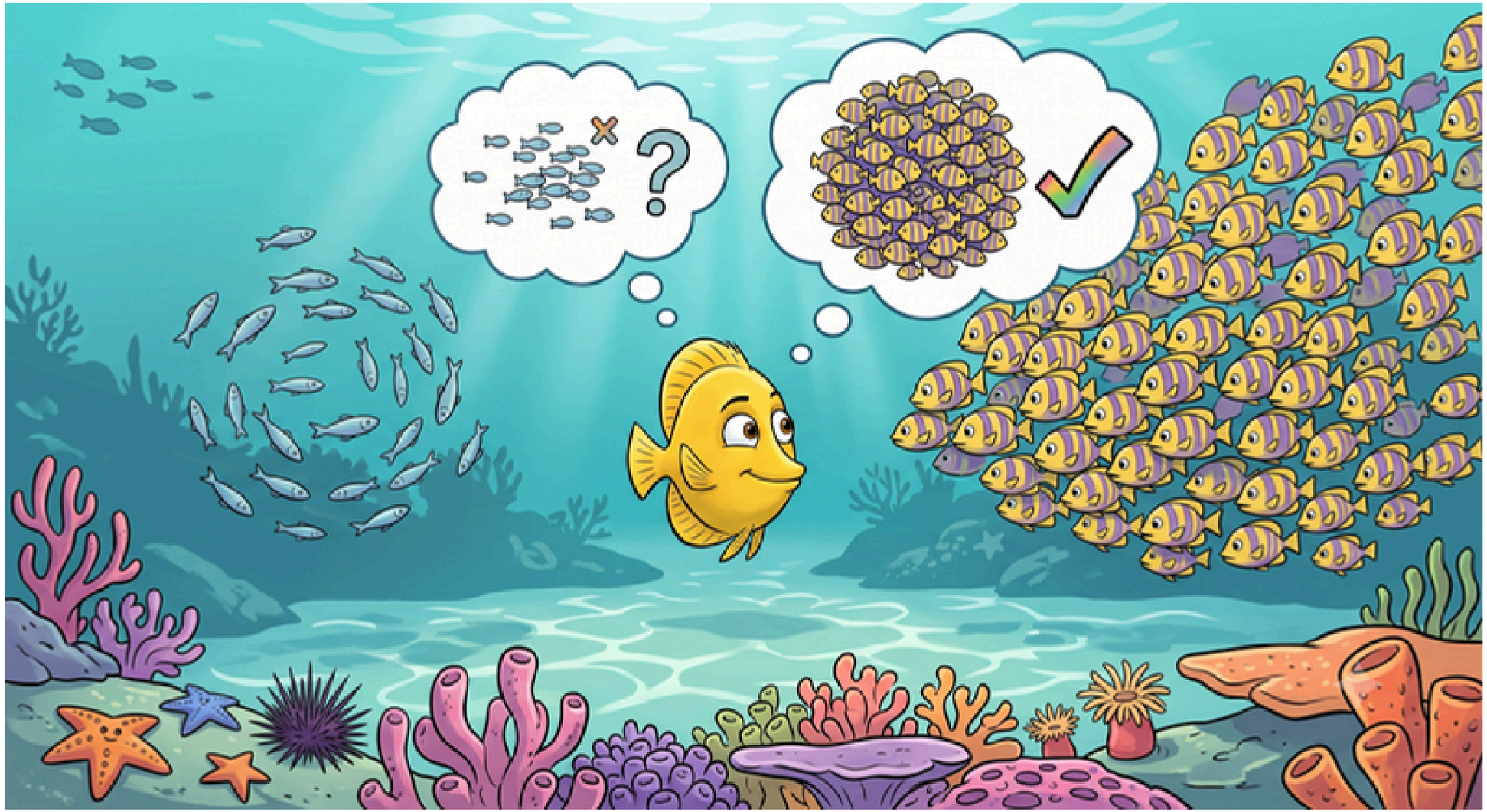
Number Representation and Data Encoding

Join the lecture through the online portal

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Do Animals need to Count?








we frogs also do need to count.
1
2
3
...
2
3

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Why Did Humans Need Numbers?

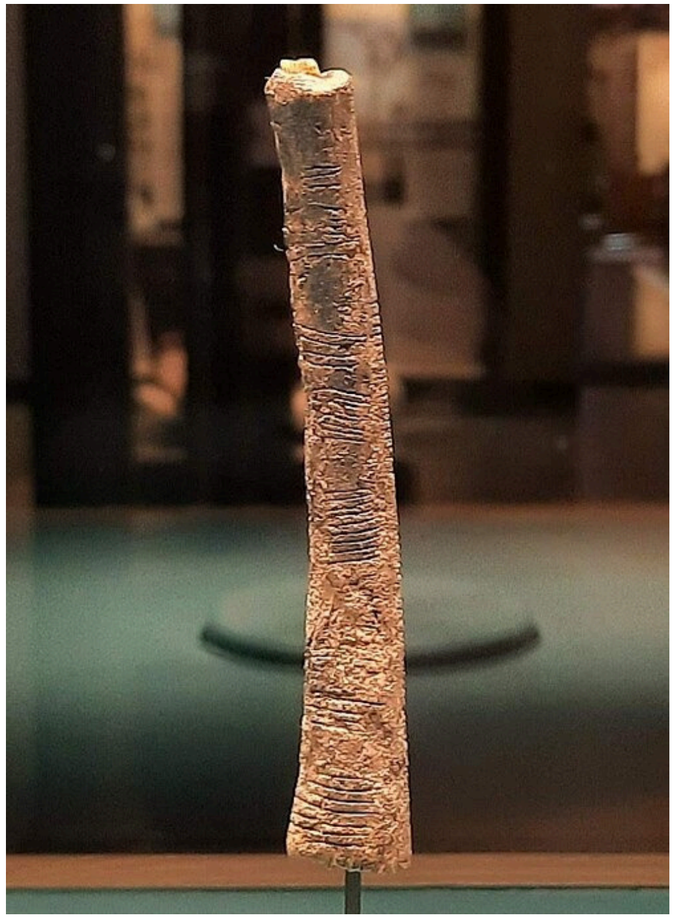
How he must have Counted?

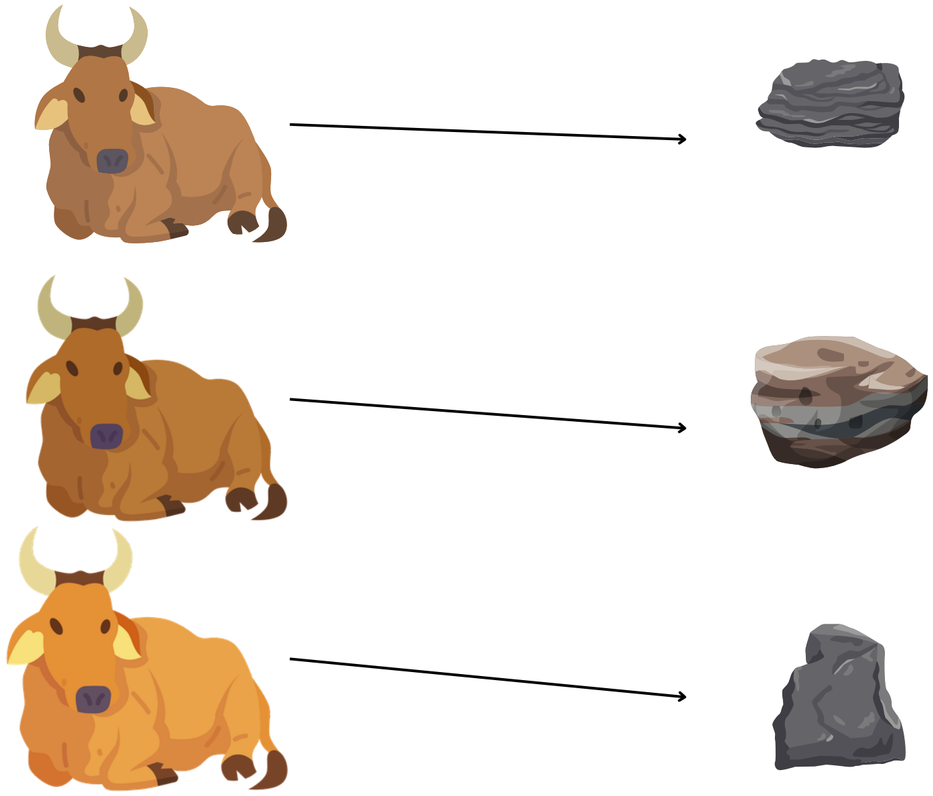


One-to-one correspondence



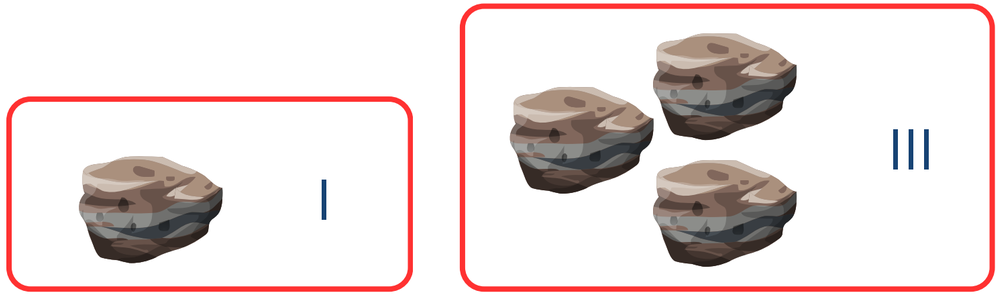
40000 years old bone with tally marks





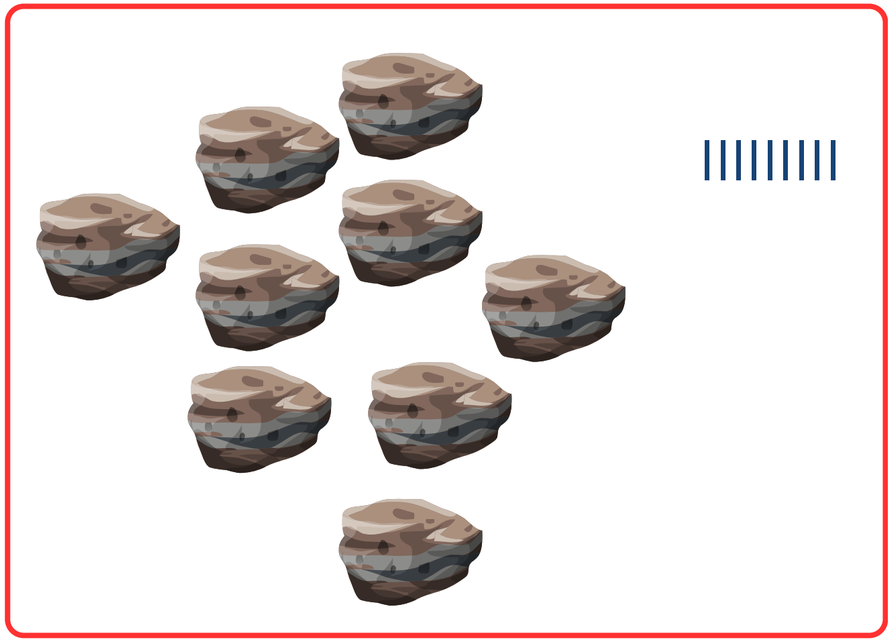
Unary Number System

One Symbol



The most basic form of counting is the **unary number system**, where **each symbol represents one quantity**. So if we want to represent 3 cows, we could use 3 symbols.

However, it becomes tedious for larger quantities.





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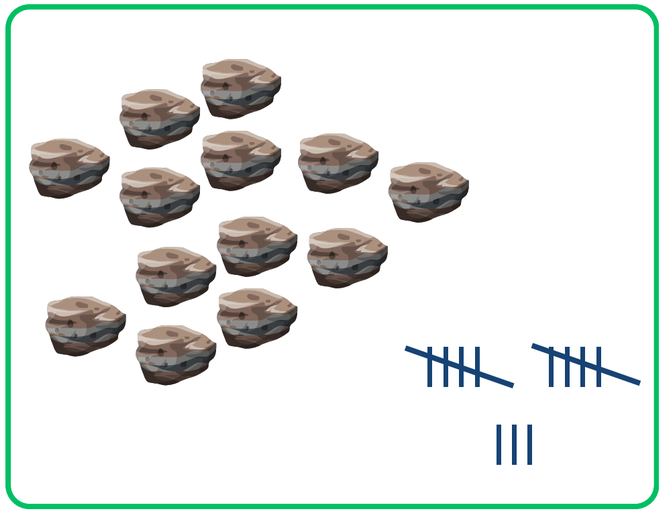
Any Limitation in Unary System?

Creating New Symbols

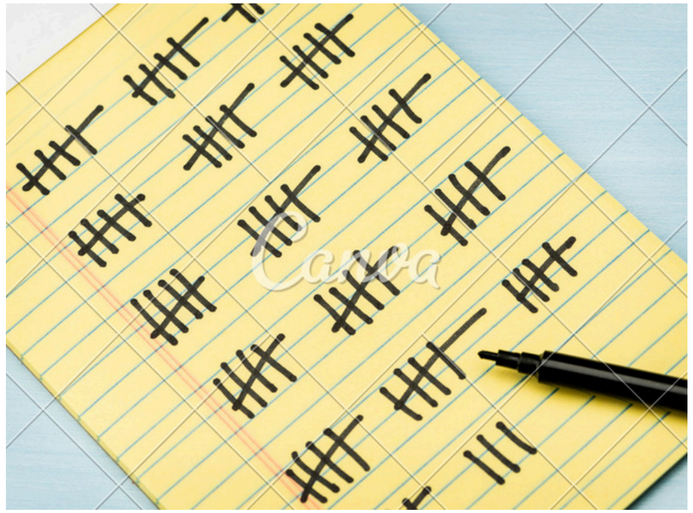
Humans developed more complex counting symbols to represent larger quantities more efficiently.

As you can see, representing a number such as 20 requires 20 symbols in unary number system, but if we decide to use a different symbol for 5, we can represent 20 with just 4 symbols.





Different symbols were created to represent different quantities.



Tally Marks



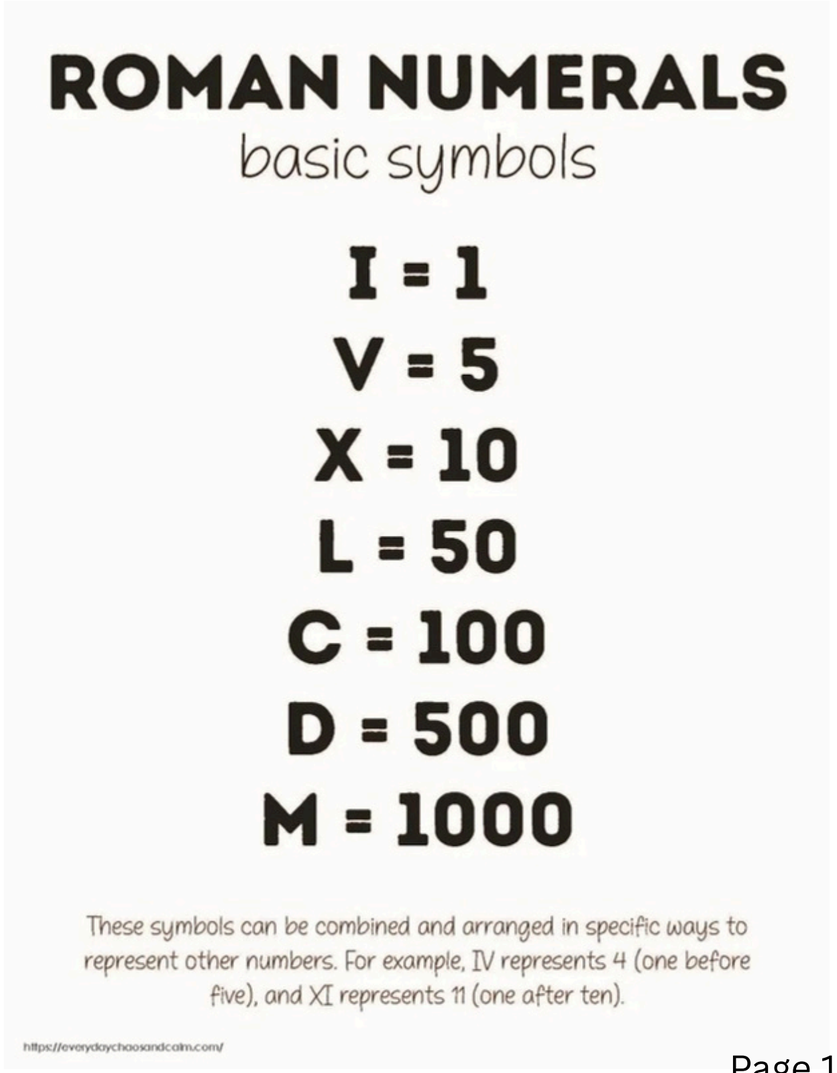
Roman Numbers



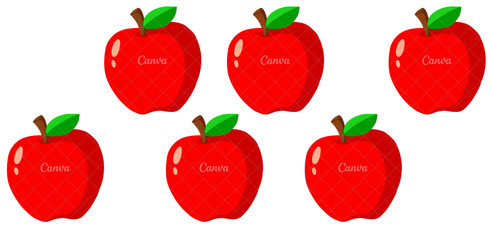
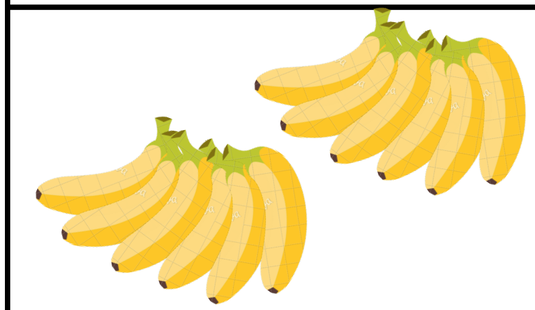
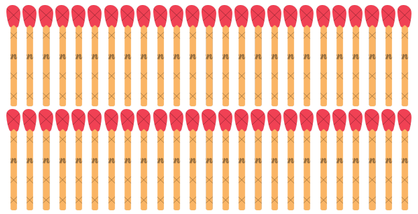
Hieroglyphics

Roman Number System

Non-Positional
Additive



Examples

Items	Unary	Roman	Decimal
	IIIIII	VI	6
	IIIIIIIIIIII	XII	12
	IIIIIIIIII

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Is Quantity changing or Just Representation?

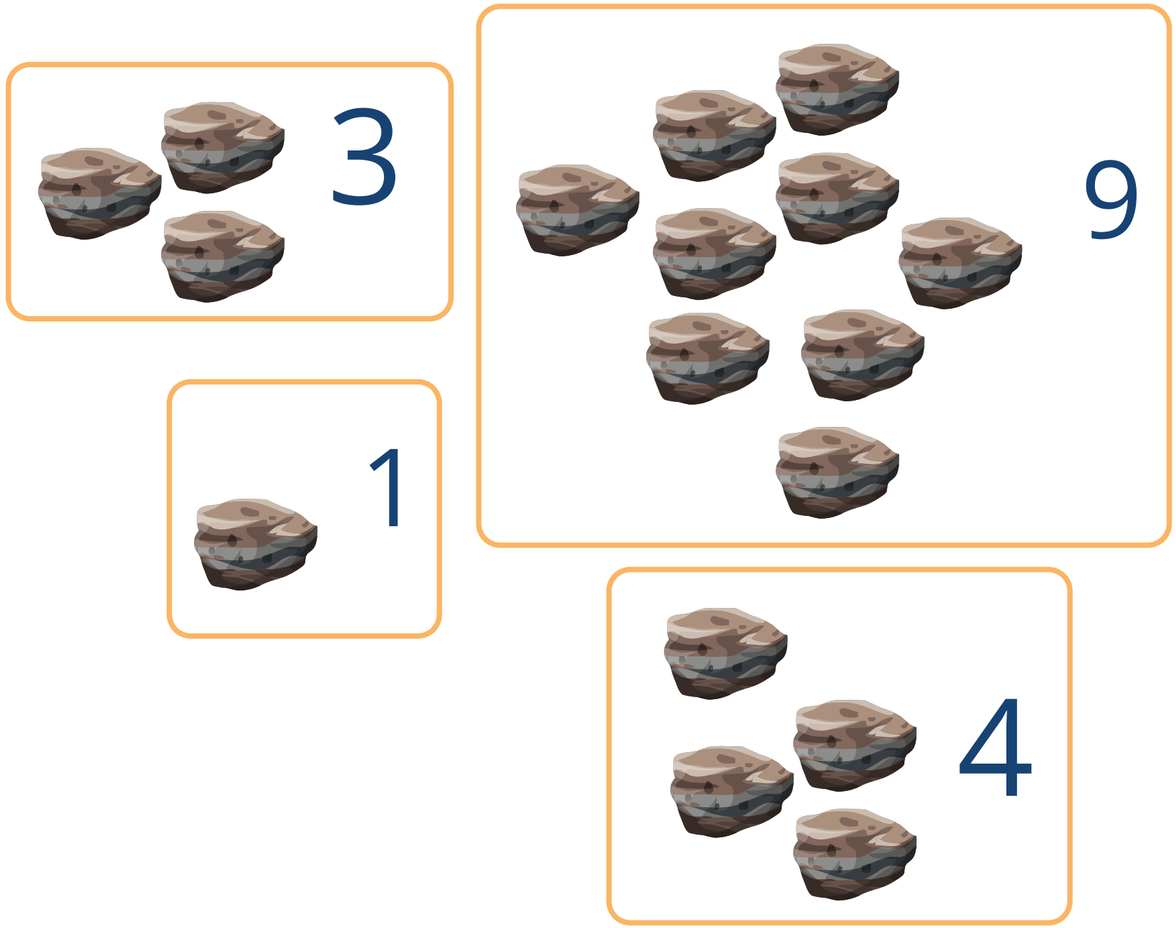
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Limitations of Roman Numerals?

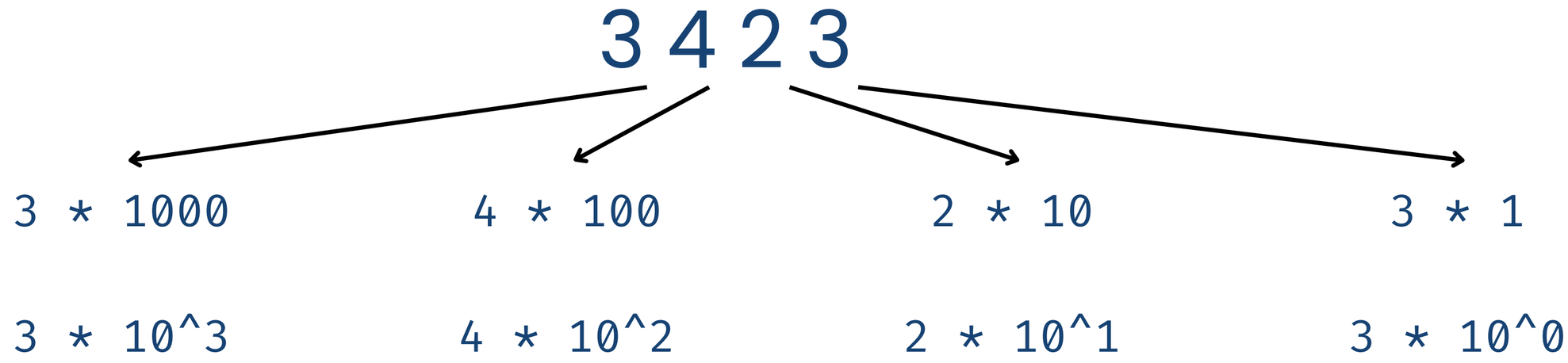
Decimal Number System

The decimal number system, also known as **base-10**, uses **ten digits (0-9)** to represent all numbers.

Positional System
Introduction of Zero



Positional Number System

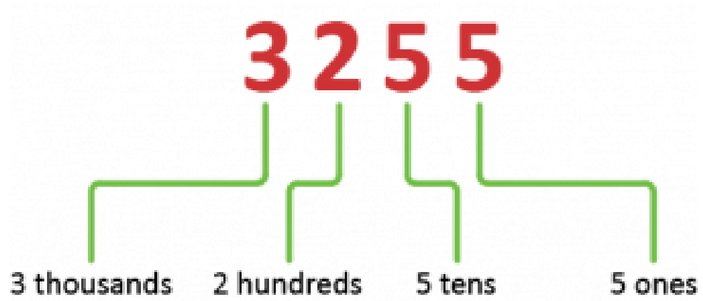


When the value of a digit depends on its position within the number, it is called a **positional number system**.

Decimal system is a positional number system, where each time we move left, the value is multiplied by **10 times** the previous one.

Grouping & Place Value

```
 0                zero ones
 1                 one ones
 2                 two ones
 3               three ones
 4                four ones
 5                five ones
 6                 six ones
 7               seven ones
 8               eight ones
 9                nine ones
10     one tens, zero ones
```



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Roman vs Decimal System

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Create a base-5 counting system and show the symbols for first 15 numbers.

Allowed Symbols 0, 1, 2, 3, 4

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Create a base-5 counting system and show the symbols for first 15 numbers.

```
0 - 0
1 - 1
2 - 2
3 - 3
4 - 4
5 - 10
6 - 11
7 - 12
8 - 13
9 - 14
10 - 20
11 - 21
12 - 22
13 - 23
14 - 24
```

There are 10 types of people in the world: those who understand binary, and those who don't

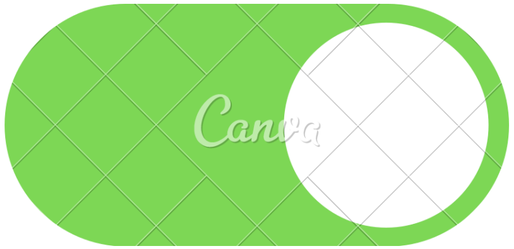
Binary Number System

Two Symbols

As the most basic electronic devices can have 2 states, on or off, they can be represented very easily using binary numbers 0 and 1.

Hence, the binary numbers play a critical role in design of digital electronic systems.

1



0



How We Count in Binary

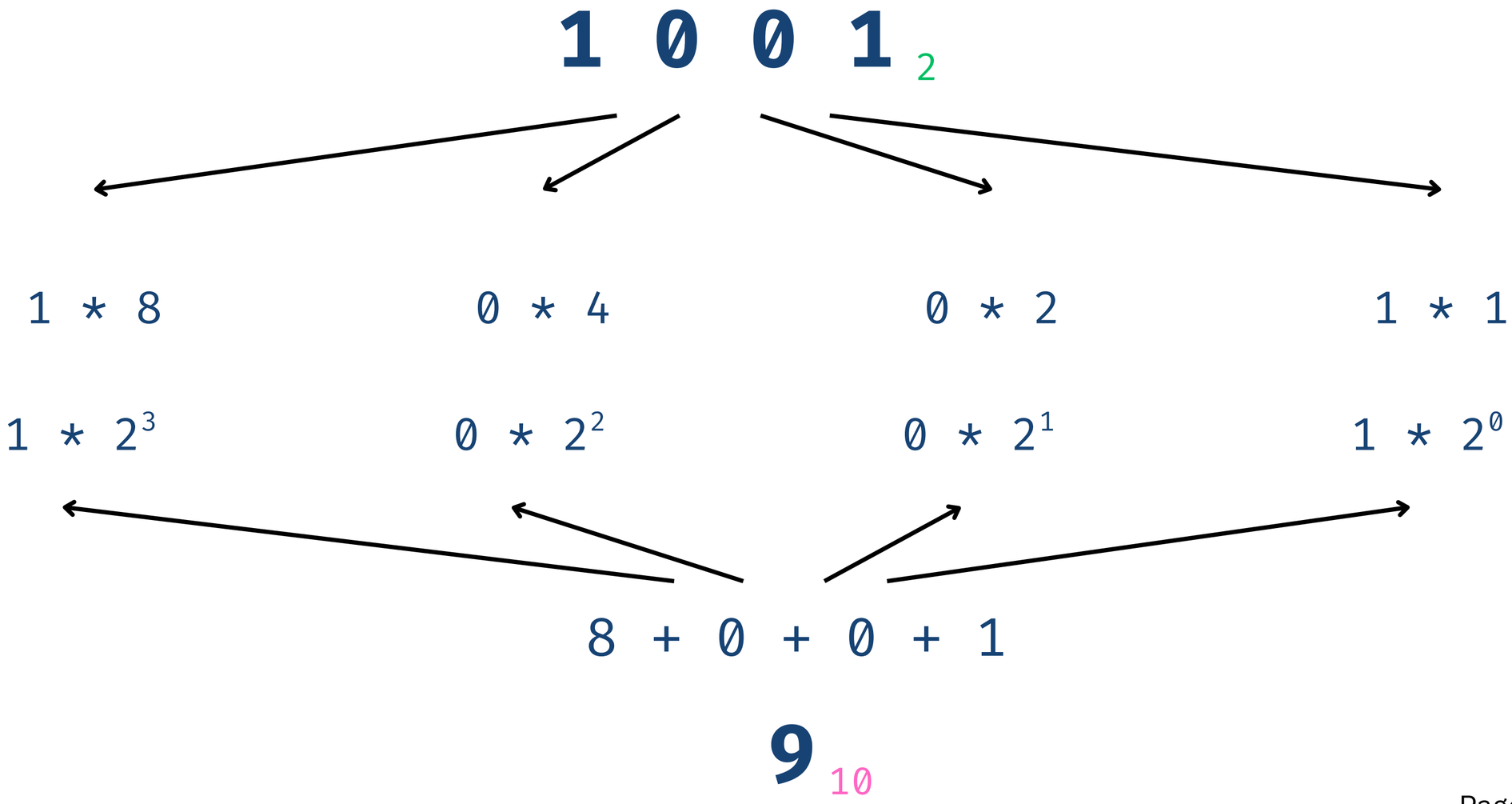
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English		Decimal		Binary
zero	→	0_{10}	→	0_2
one	→	1_{10}	→	1_2
two	→	2_{10}	→	10_2
three	→	3_{10}	→	11_2
four	→	4_{10}	→	100_2
five	→	5_{10}	→	101_2

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Binary & Decimal Conversions?

Binary to Decimal



Decimal to Binary

Step 1: Divide the given number **13** repeatediy by 2 until you get '0' as the quotient

$13 \div 2 = 6$ (Remainder 1)

$6 \div 2 = 3$ (Remainder 0)

$3 \div 2 = 1$ (Remainder 1)

$1 \div 2 = 0$ (Remainder 1)

Step 2: Write the remainders in the reverse order

1 1 0 1

$\therefore 13_{10} = 1101_2$

(Decimal) (Binary)

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Every positive integer can be written uniquely as a sum of powers of any base b≥2 with coefficients from 0..b−1. This is not a convention.

this theorem is a property of numbers themselves.